Department of Electrical Engineering
College of Engineering
German International University in Cairo

ENRB 603 (Spring 2025) Project (1)

Title: *Stepper Motor Control*

Score: / 100

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Tutorial:P8

Due on: May 20th, 2025

- **Problem:**

It is required to design and implement a motion control system for a stepper motor, using PLC and a suitable driver (optional), as illustrated in the figure. Both speed, rotation direction, and position need to be controlled.

(a)

(b)

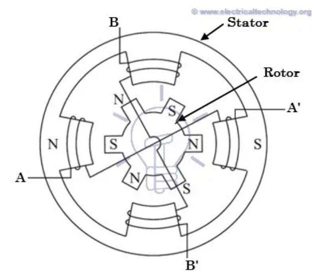
(c)

Task	Score	Comments
1	/15	
2	/20	
3	/50	

4	/15	

- **Student's feedback:**

Stepper Motor Control



Operation of stepper motors

A stepper motor is a brushless motor that has permanent magnet poles that offer a natural stopping point for the motor shaft. This means that the poles of the motor stop and position themselves at precise angles in an open loop system and no feedback is required. Stator windings are energized resulting in a magnetic field that rotates the shaft. Coils of wire are static in the center and the permanent magnets spin around them on the outside.The stepper motor is operated by having a programmable controller (pulse generator) generate pulses that are input to a driver, which in turn supplies the drive current to the motor.

Adjusting Speed of Motor

The speed of a stepper motor is proportional to the rate of input pulses. A higher pulse rate (higher pulse frequency) causes the speed of stepper motor rotation to increase proportionally

Direction of Rotation

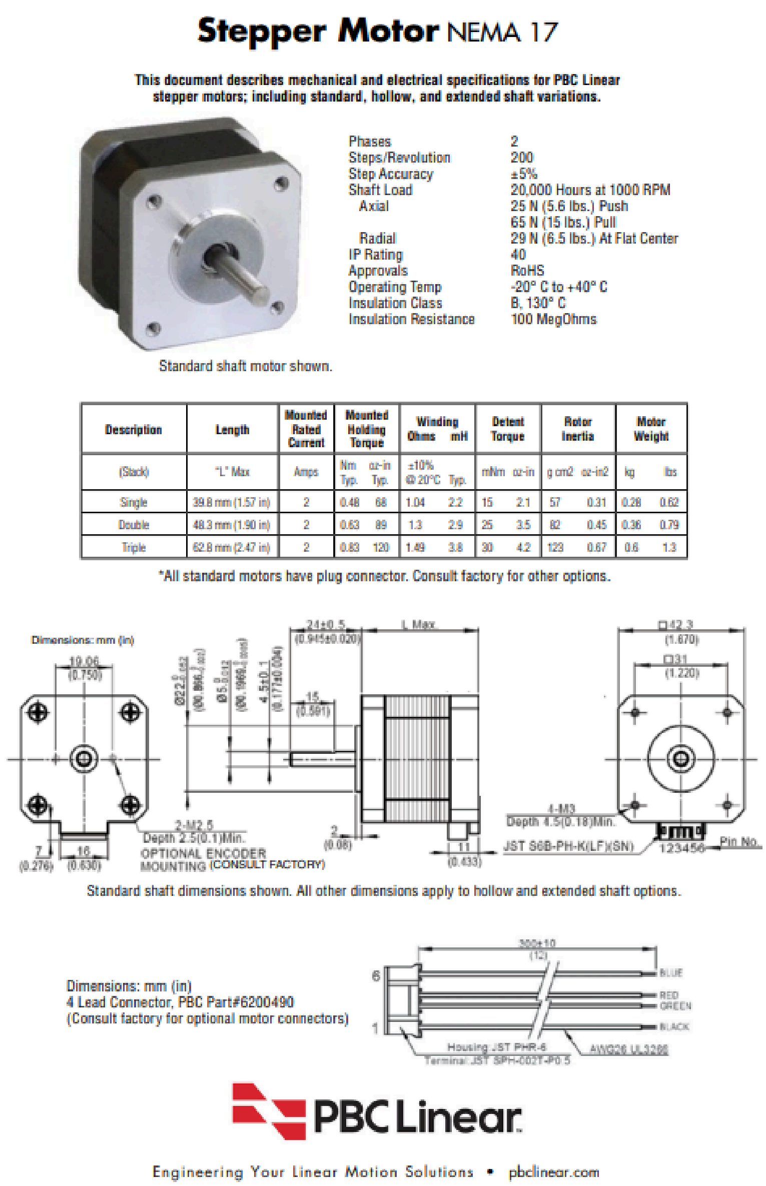
The stepper motor driver has a DIR pin which reverses the current flow through each coil in the stepping sequence for the motor, which causes the magnetic field to rotate the other way reversing the motor direction. To modify the direction in which the stepper motor moves, the amount of voltage being sent through the DIR pin is changed. If the user wants the motor to go the clockwise direction then they will pass high voltage (5V) through the pin, as for anticlockwise, the user will have to pass low voltage (0V) through the pin

Position Control

Stepper motor rotation is proportional to the number of input pulses, and this enables accurate positioning. The incremental step size of a stepper motor is fixed to a certain degree of rotation depending on the number of electromagnetic poles (the greater the number of poles, the smaller the least possible change/step is thus increasing positioning accuracy). More pulses sent to the motor means more steps and bigger angle rotated

Motor and Driver

The NEMA 17 motor will be used. The TB6600 stepper motor driver will be used to connect the PLC to the stepper motor. As stated above, it drives current to the motor to rotate and provides the ability to control the direction of rotation so it is essential. It also allows full step and half step control.



Stepper Motor NEMA 17

This document describes mechanical and electrical specifications for PBC Linear stepper motors; including standard, hollow, and extended shaft variations.

Phases	2
Steps/Revolution	200
Step Accuracy	±5%
Shaft Load	20,000 Hours at 1000 RPM
Axial	25 N (5.6 lbs.) Push 65 N (15 lbs.) Pull
Radial	29 N (6.5 lbs.) At Flat Center
IP Rating	40
Approvals	RoHS
Operating Temp	-20° C to +40° C
Insulation Class	B, 130° C
Insulation Resistance	100 MegOhms

Standard shaft motor shown.

Description	Length	Mounted Rated Current	Mounted Holding Torque		Winding Ohms	mH	Detent Torque		Rotor Inertia		Motor Weight	
(Stack)	"L" Max	Amps	Nm Typ.	oz-in Typ.	±10% @ 20°C	Typ.	mNm	oz-in	g cm2	oz-in2	kg	lbs
Single	39.8 mm (1.57 in)	2	0.48	68	1.04	2.2	15	2.1	57	0.31	0.28	0.62
Double	48.3 mm (1.90 in)	2	0.63	89	1.3	2.9	25	3.5	82	0.45	0.36	0.79
Triple	62.8 mm (2.47 in)	2	0.83	120	1.49	3.8	30	4.2	123	0.67	0.6	1.3

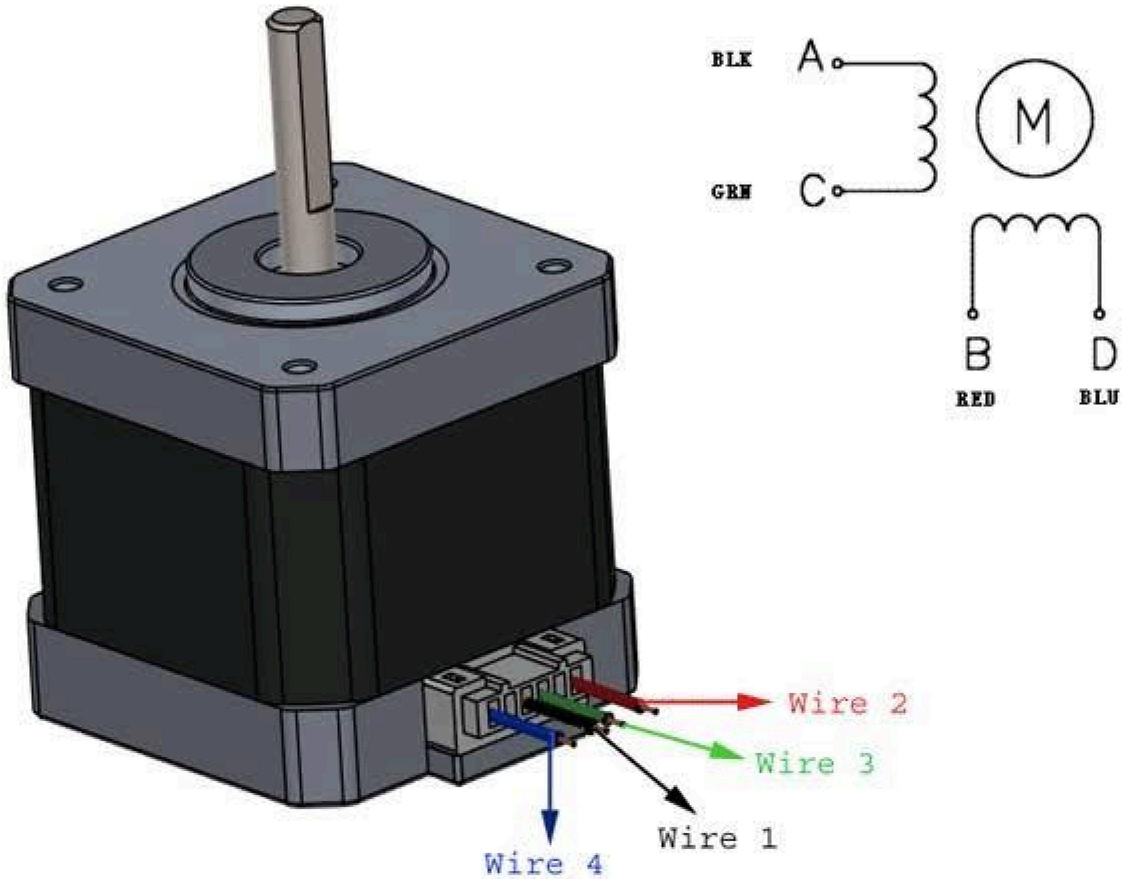
*All standard motors have plug connector. Consult factory for other options.

Standard shaft dimensions shown. All other dimensions apply to hollow and extended shaft options.

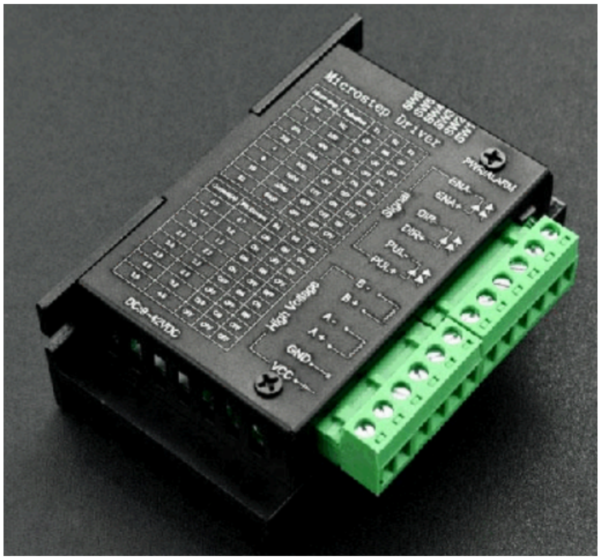
Dimensions: mm (in)
4 Lead Connector, PBC Part#6200490
(Consult factory for optional motor connectors)

PBC Linear

Engineering Your Linear Motion Solutions • pbclinear.com https://pages.pbclinear.com/rs/909-BFY-775/images/Data-Sheet-Stepper-Motor-Support.pdf



https://how2electronics.com/control-stepper-motor-with-drv8825-raspberry-pi-pico/



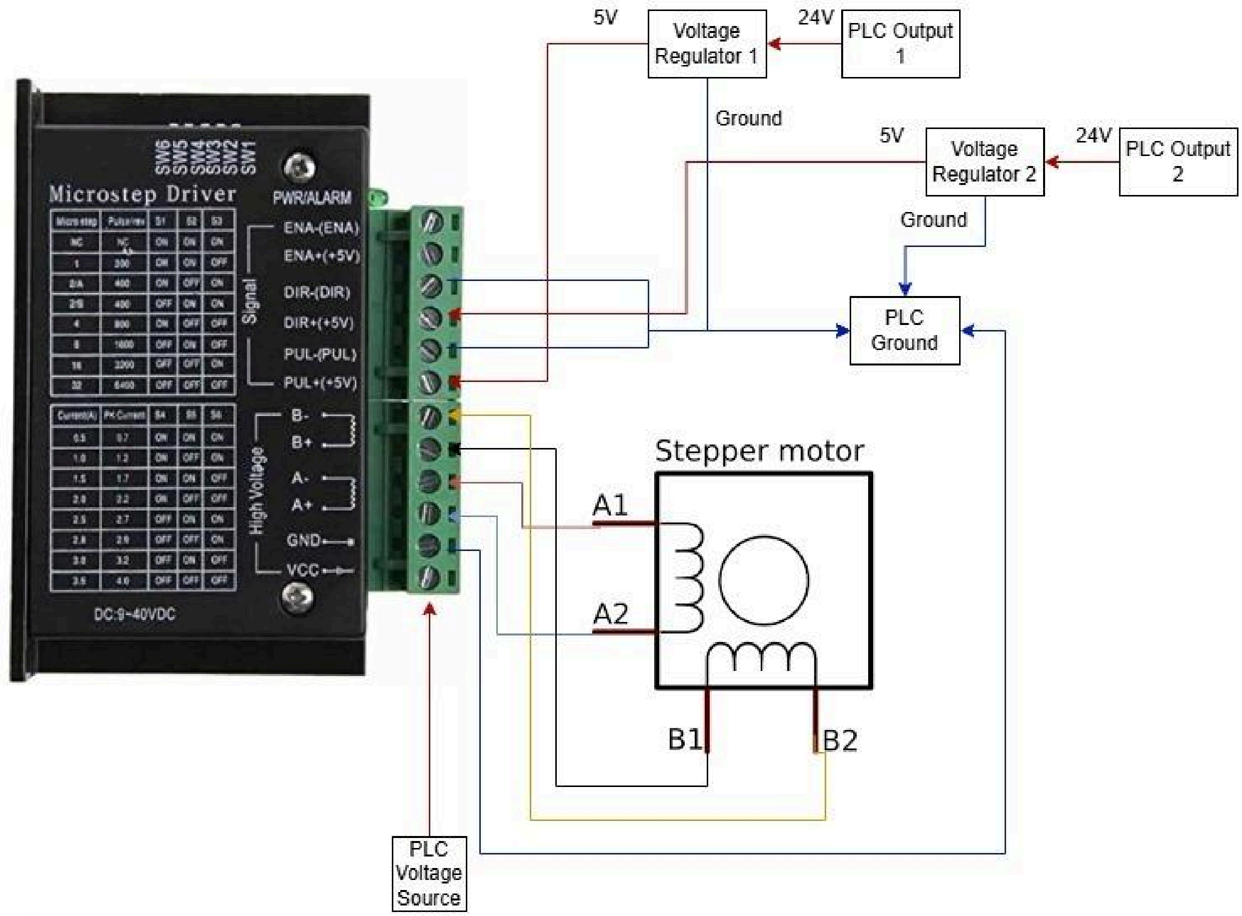
SPECIFICATION

- Input Current: 0~5A
- Output Current: 0.5-4.0A
- Control Signal: 3.3~24V
- Power (MAX): 160W
- Micro Step: 1, 2/A, 2/B, 4, 8, 16, 32
- Temperature: -10 ~ 45°C
- Humidity: No Condensation
- Dimension: 96*56*33 mm/ 3.78*2.2*1.3 inches
- Weight: 0.2 kg
- Drive IC: TB67S109AFTG

https://www.alldatasheet.com/html-pdf/1759229/DFROBOT/TB6600/341/1/TB6600.html

Wiring Diagram

This figure will demonstrate all the connections that we have used to make this stepper motor work.



Truth Tables

The following truth tables will discuss how each of the full step and half step will work in terms of which coils are excited and which coils are dormant and how a step occurs.

Full step: -

Clockwise Truth Table				
Step	**A**	**B**	**A-**	**B-**

1	1	0	1	0
2	0	1	1	0
3	0	1	0	1
4	1	0	0	1

Counter-Clockwise Truth Table				
Step	**A**	**B**	**A-**	**B-**
1	1	0	0	1
2	0	1	0	1
3	0	1	1	0
4	1	0	1	0

Half Step: -

Clockwise Truth Table				
Step	**A**	**B**	**A-**	**B-**
1	1	0	0	0
2	1	0	1	0
3	0	0	1	0
4	0	1	1	0
5	0	1	0	0
6	0	1	0	1
7	0	0	0	1
8	1	0	0	1

Counter-Clockwise Truth Table				
Step	**A**	**B**	**A-**	**B-**
1	1	0	0	1
2	0	0	0	1
3	0	1	0	1
4	0	1	0	0

5	0	1	1	0
6	0	0	1	0
7	1	0	1	0
8	1	0	0	0

Further Info about the Stepper motor

As seen from the above truth tables, the full step only requires 4 steps while the half step requires 8 steps. So by deciding upon the angle size we can deduce that

Step Type	Step Angle (Preset from manufacturer)	90° Rotation	180° Rotation	Full Turn (360°)
Full-step	1.8°	50 steps	100 steps	200 steps
Half-step	0.9°	100 steps	200 steps	400 steps

Steps required=Target angle/Step angle

For the speed of the motor, it has nothing to do with the pulse themselves, and more about the delay between each pulse. As the delay decreases, the speed of the motor would increase.

This is for Speed 2 > Speed 1.

Connection explanation: -

To achieve precise motion control, the stepper motor is driven by selectively energizing its coils in a sequence dictated by the truth table (referenced earlier). Depending on the required step resolution—full-step, half-step, or microstepping—the coils are activated in a specific order:

1. Coil Excitation Sequence

For each step, one or multiple coils are powered simultaneously. After a brief hold time, the coils are depowered before switching to the next set in the sequence. This creates the magnetic field rotation necessary for motor movement.

2. PLC Control Logic

Direction Control: Digital output Q0.0 determines rotation direction:

0 → Maintain current direction.

1 → Reverse rotation.

Step Trigger: Output Q0.1 sends a pulse signal to the motor driver (TB6600), commanding a single step per rising edge.

3. Voltage Regulation & Power Handling

The PLC outputs 24V logic, but the TB6600 accepts only 5V input signals.

A voltage regulator (e.g., voltage divider or optocoupler) ensures signal compatibility.

The TB6600 driver is powered directly by the PLC’s 24V supply, as it supports this voltage for its power input.

Code explanation

First block: MC_Power

- EN (Enable): Activated by "Tag_1" (M0.0)
- Axis_1: The axis being controlled (motor)
- Enable: Set to 1 (true) to enable the axis

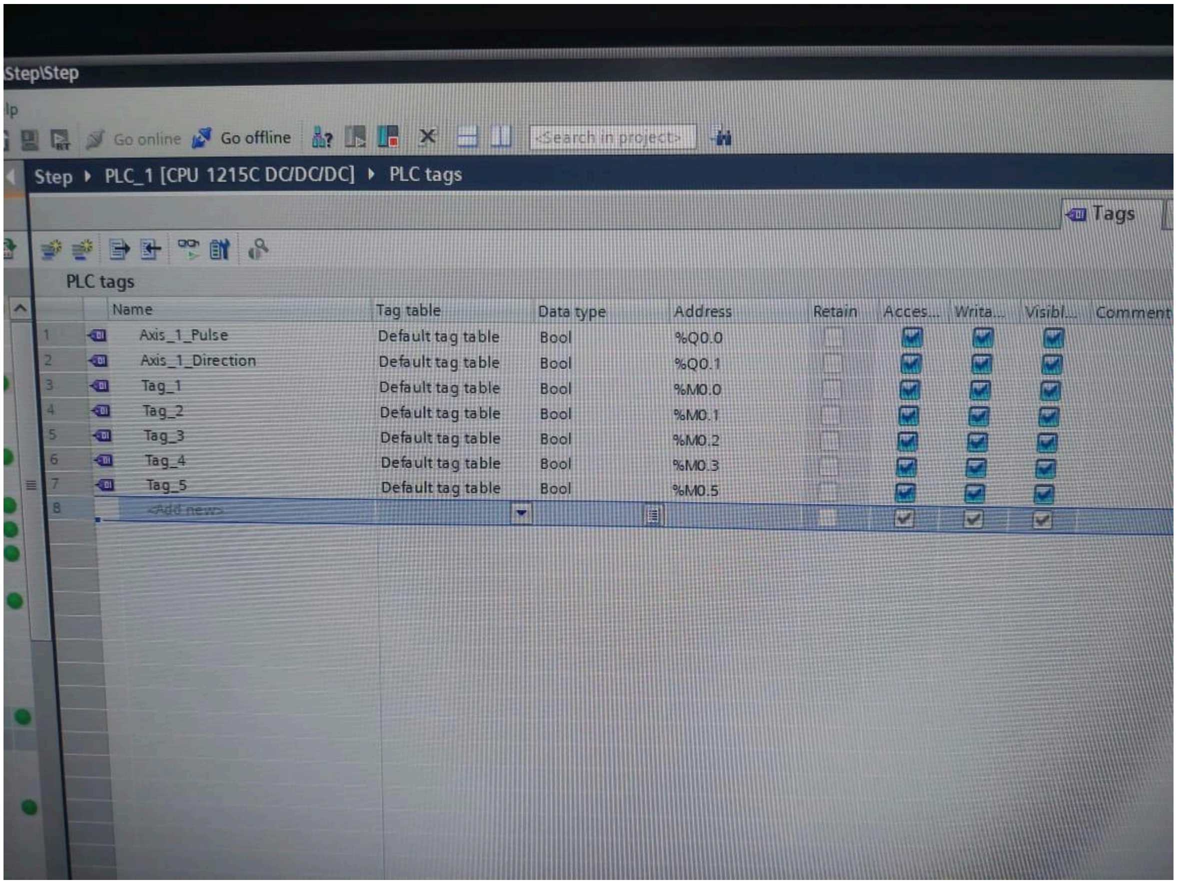
Second block: MC_MoveJog

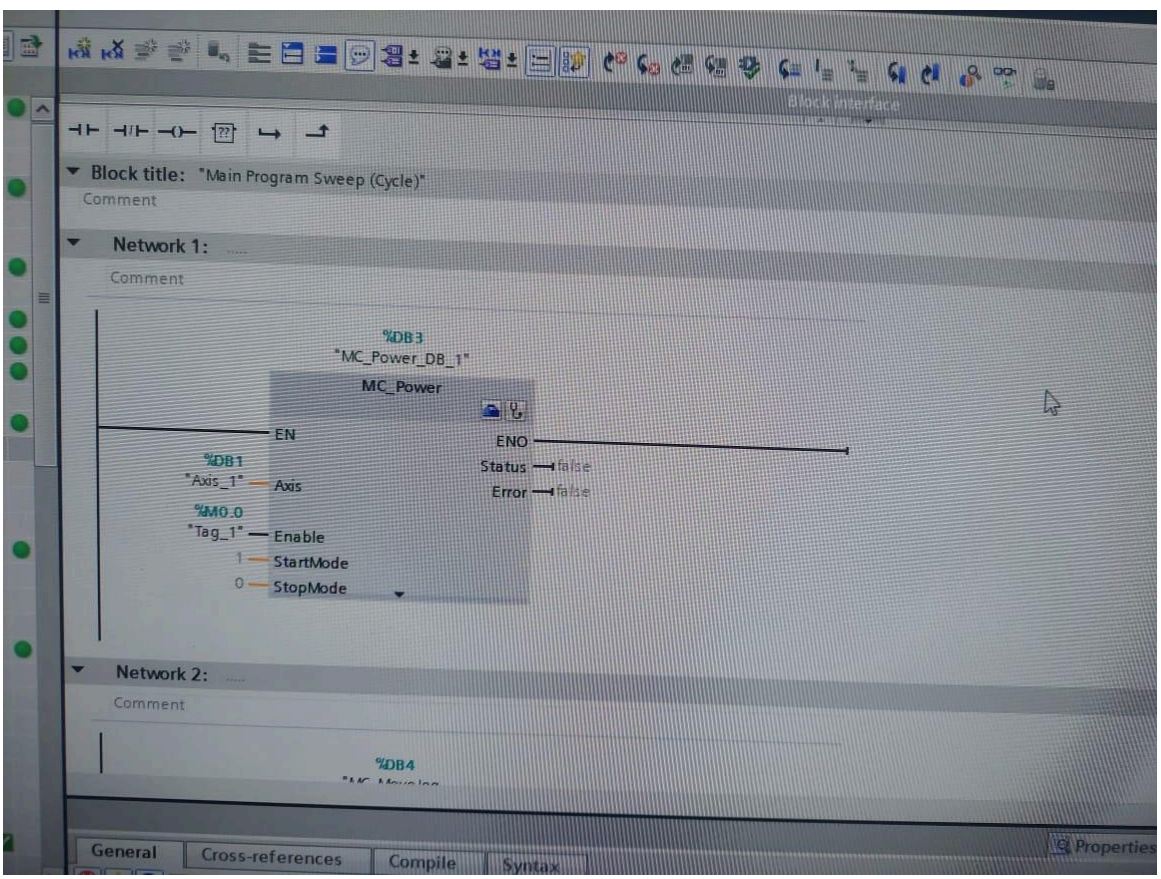
- Axis_1: The axis being controlled
- JogForward: Activated by "Tag_3" (M0.2) which makes the motor rotate
- JogBackward: Activated by "Tag_4" (M0.3) which makes the motor rotate in the opposite direction
- Velocity: Set to 200.0 (speed of movement)

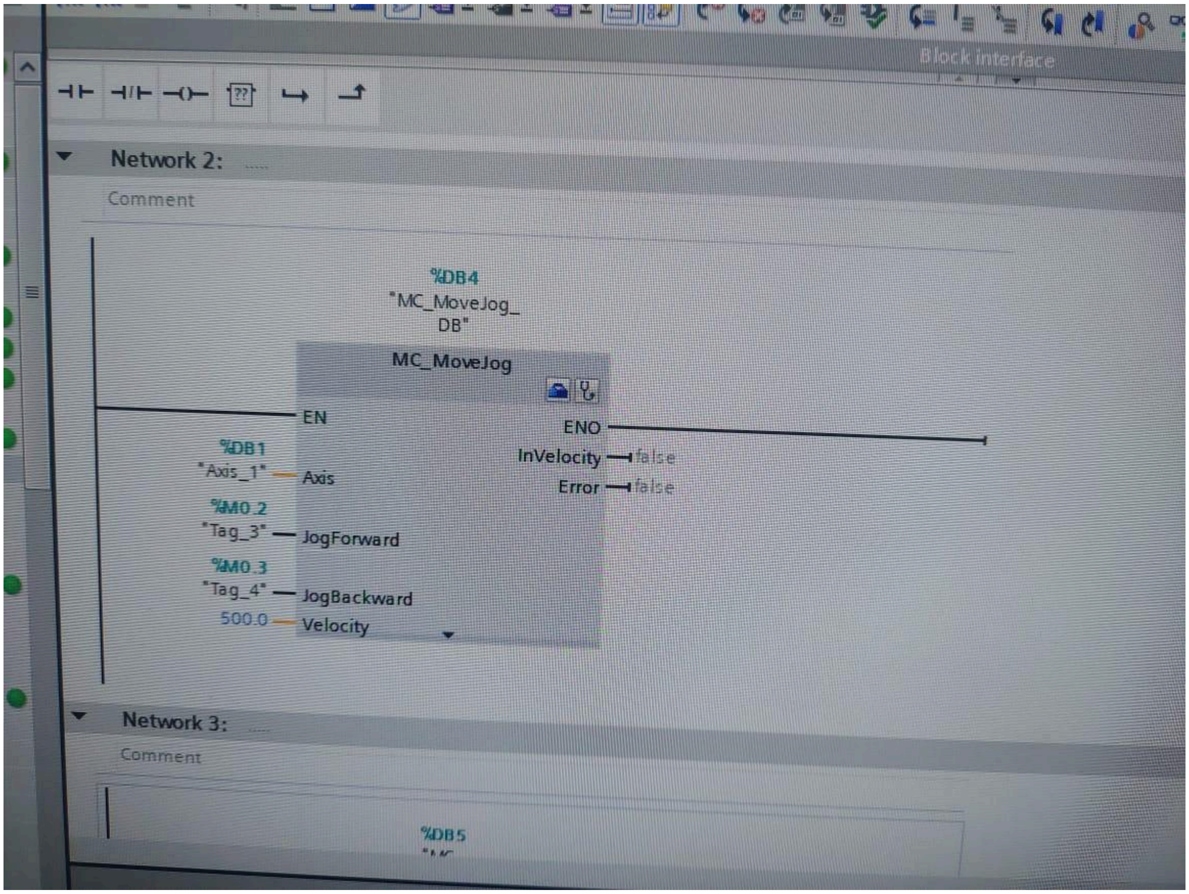
Third Block: MC_MoveRelative

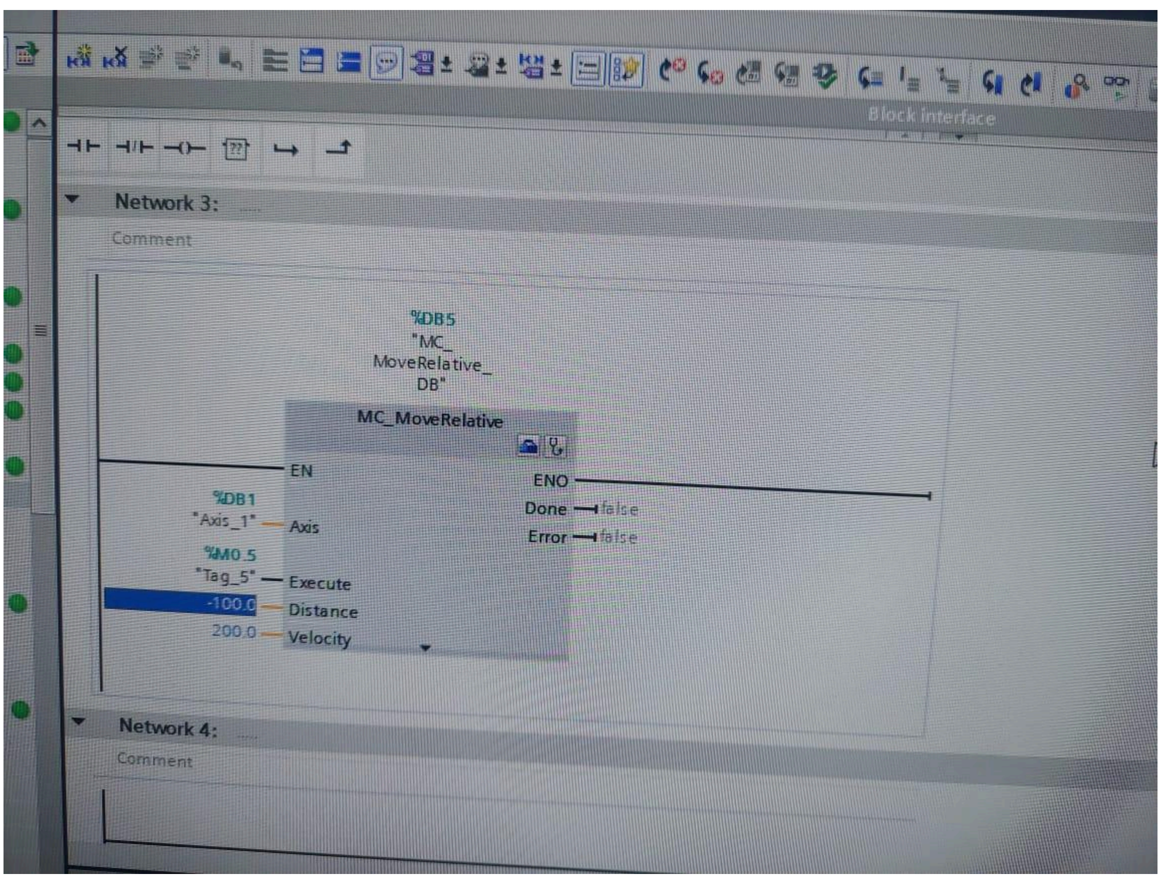
- Axis_1: The axis being controlled
- Execute: Activated by "Tag_5" (M0.5) moves the motor for the given distance at the given speed
- Distance: 100 (distance of rotation)
- Velocity: 200.0 (speed of movement)

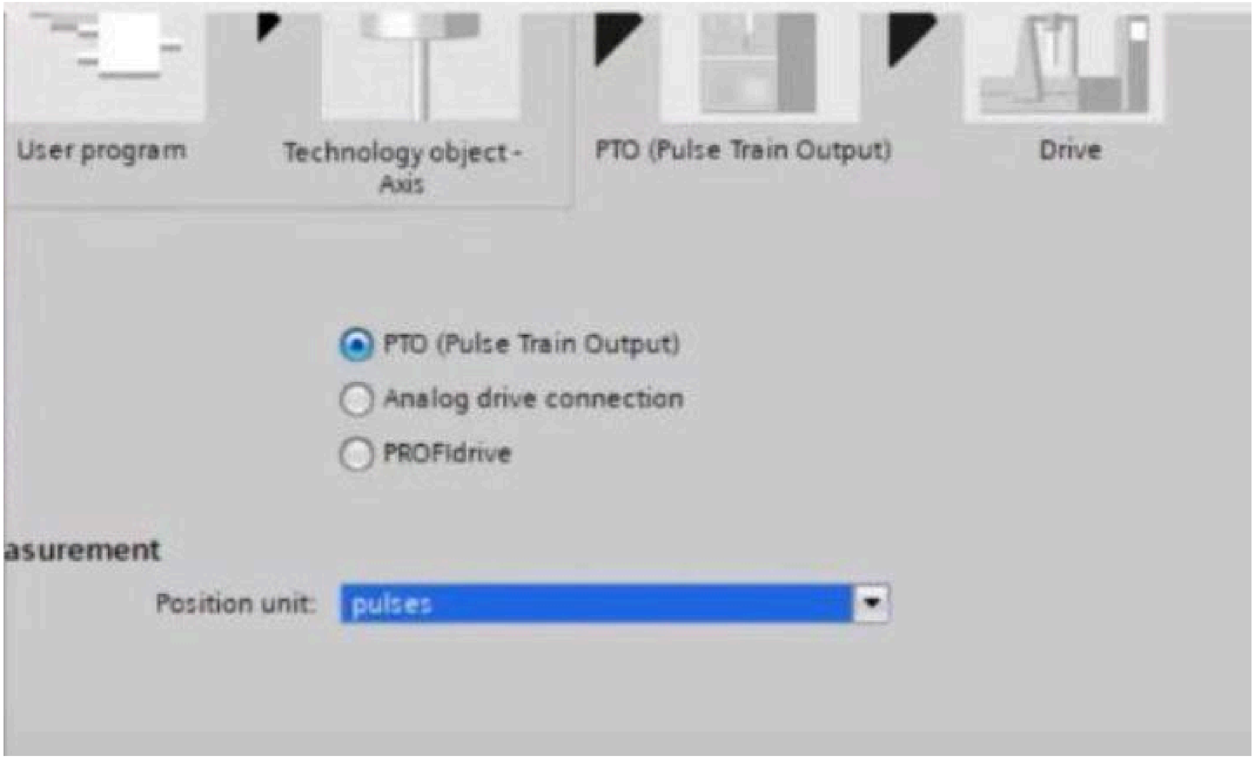
  (example: 100/200 = making half a revolution)

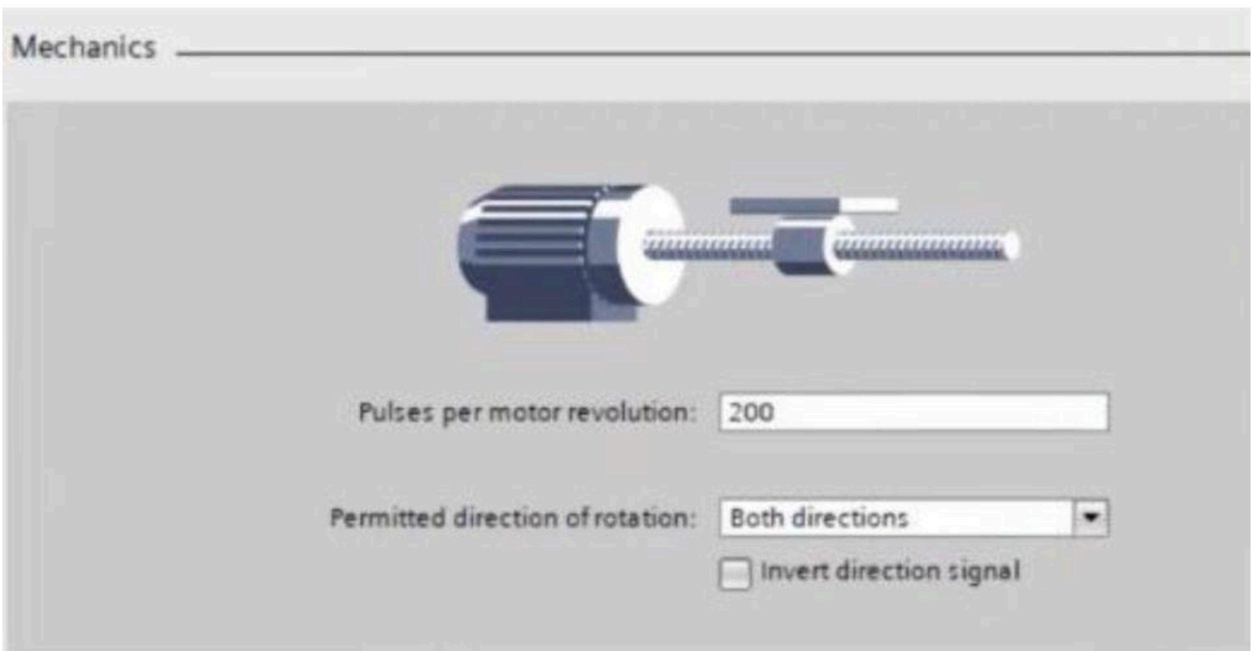

Step\Step
Go online
Go offline
Step ▸ PLC_1 [CPU 1215C DC/DC/DC] ▸ PLC tags
Tags
PLC tags
Name
Tag table
Data type
Address
Retain
Acces...
Writa...
Visibl...
Comment
1 Axis_1_Pulse Default tag table Bool %Q0.0
2 Axis_1_Direction Default tag table Bool %Q0.1
3 Tag_1 Default tag table Bool %M0.0
4 Tag_2 Default tag table Bool %M0.1
5 Tag_3 Default tag table Bool %M0.2
6 Tag_4 Default tag table Bool %M0.3
7 Tag_5 Default tag table Bool %M0.5
8 <Add new>


Block title: "Main Program Sweep (Cycle)"
Comment
Network 1:
Comment
%DB3
"MC_Power_DB_1"
MC_Power
EN
ENO
Status
Error
%DB1
"Axis_1"
Axis
%M0.0
"Tag_1"
Enable
1
StartMode
0
StopMode
Network 2:
Comment
%DB4
General
Cross-references
Compile
Syntax
Properties


Block interface
Network 2:
Comment
%DB4
"MC_MoveJog_DB"
MC_MoveJog
EN
ENO
%DB1
"Axis_1"
Axis
InVelocity
false
Error
false
%M0.2
"Tag_3"
JogForward
%M0.3
"Tag_4"
JogBackward
500.0
Velocity
Network 3:
Comment
%DB5


Block interface
Network 3:
Comment
%DB5
"MC_MoveRelative_DB"
MC_MoveRelative
EN
ENO
%DB1
"Axis_1"
Axis
Done
false
Error
false
%M0.5
"Tag_5"
Execute
-100.0
Distance
200.0
Velocity
Network 4:
Comment

- 
User program
Technology object - Axis
PTO (Pulse Train Output)
Drive
PTO (Pulse Train Output)
Analog drive connection
PROFIdrive
asurement
Position unit:
pulses


Mechanics
Pulses per motor revolution:
200
Permitted direction of rotation:
Both directions
Invert direction signal